IR fixed points and RG phases of $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories in the non-Veneziano large-N limit

April 23, 2026

Abstract

We classify four-dimensional $\mathcal{N} = 1$ asymptotically free $\mathrm{SU}(N)_1 \times \mathrm{SU}(N)_2$ quiver gauge theories admitting a non-trivial large-N limit without Veneziano scaling of fundamental matter, and analyse the superconformal fixed points and renormalisation-group flow phases that arise in the two-coupling plane. Seven universality classes emerge, all satisfying $a = c$ to leading order in $1/N$, each with multiple distinct quiver representatives. Applying the boundary analysis of Intriligator–Wecht (`hep-th/0502049`) to both axis SCFTs, we compute the boundary one-loop coefficients $b_0^{(a)}(g_b^\star)$ and classify each class by the sign pattern of their leading-N coefficients, obtaining the flow topology in the (g_1, g_2)-plane for every class. As our main result, we exhibit an explicit realisation of the "both axes IR-stable" topology (Figure 6 of [1]), which those authors noted they could not construct: deforming class 16 by N_f fundamentals on node 2 in the window $0.427\,N - 1 < N_f < N - 1$ yields precisely this phase.

Contents

1 Introduction **2**

2 Setup: two-node quivers in the non-Veneziano large-N limit **3**
- 2.1 Field content and gauge group . . . 3
- 2.2 Anomaly cancellation and the one-loop beta function . . . 4
- 2.3 The non-Veneziano large-N limit and the bound $k_{\mathrm{tot}} \leq 4$ . . . 5
- 2.4 IR fixed points and a-maximisation . . . 6
- 2.5 Axis SCFTs and the boundary one-loop coefficient $b_0^{(a)}(g_b^\star)$ . . . 6
- 2.6 Database and scope . . . 6

3 Classification via a-maximisation **7**
- 3.1 Enumeration pipeline . . . 7
- 3.2 The seven non-Veneziano equal-rank classes . . . 7
- 3.3 Multiple representatives per class . . . 8
- 3.4 Representative R-charges . . . 8
- 3.5 The two special classes: 1 and 44 . . . 9

4 Boundary analysis **10**
- 4.1 The boundary one-loop coefficient $b_0^{(b)}(g_a^\star)$ . . . 10
- 4.2 The seven equal-rank classes . . . 10
- 4.3 RG-topology classification of the seven classes . . . 10
- 4.4 Approximation: interior vs. boundary R_{bif} . . . 11

5 RG flow phases and a realisation of the Intriligator–Wecht Figure 6 topology **11**
5.1 Sign pattern of $(\alpha_1^{(2\star)}, \alpha_2^{(1\star)})$ and the RG topology . . . 11
5.2 Class-by-class phase analysis . . . 12
5.3 Summary of equal-rank phase distribution . . . 14
5.4 Figure 6 realisation: class 16 deformed by N_f fundamentals on node 2 . . . 14
5.5 Other classes under analogous deformations . . . 15
5.6 Remarks on the interior fixed point . . . 16

6 Discussion **16**
6.1 Summary . . . 16
6.2 Implications . . . 17
6.3 Future work . . . 17

A Mixed-rank $\mathrm{SU}(m_0N) \times \mathrm{SU}(m_1N)$ classes **18**

1 Introduction

Four-dimensional $\mathcal{N} = 1$ supersymmetric gauge theories provide one of the richest arenas for exact non-perturbative statements about renormalisation-group flow. For a single gauge group, the a-theorem [2], the conformal window of SQCD [3], and the identification of the superconformal R-symmetry by a-maximisation [4] together give a remarkably complete picture of the space of IR superconformal fixed points. Moving from a single gauge group to a quiver of several gauge factors brings two new features of immediate physical interest. First, the infra-red is no longer a single point but in general a continuous family of superconformal theories labelled by the exactly marginal ratios of the gauge couplings of the individual factors. Second, the two-loop space between the ultraviolet Gaussian fixed point and the available infra-red fixed points organises itself into distinct *RG-flow topologies*, characterised by which fixed points are attractors, which are saddles, and how basins of attraction are arranged.

For a two-node quiver with gauge group $\mathrm{SU}(N)_1 \times \mathrm{SU}(N)_2$ and coupling plane $(g_1, g_2) \in \mathbb{R}^2_{\geq 0}$, the candidate IR fixed points are the Gaussian origin, the two single-node ("axis") SCFTs obtained by decoupling one gauge factor, and any interior two-coupling SCFT that may arise. A central question is: *which axis and interior fixed points attract generic flows, and under what conditions does the interior SCFT emerge as the global IR attractor?* The answer depends on the signs of two one-loop coefficients controlling the stability of each axis SCFT against turning on the orthogonal coupling. Following Intriligator and Wecht [1], these stability coefficients are computed by a-maximising the decoupled single-node sub-theory and substituting the resulting bifundamental R-charge into an effective one-loop formula; the result is a leading-N quantity of the schematic form $b_0^{(a)}(g_b^\star)$, representing the one-loop coefficient of node a when node $b \neq a$ sits at its axis SCFT.

[1] identified three RG-flow topologies distinguished by the sign pattern $(\operatorname{sign} b_0^{(1)}(g_2^\star), \operatorname{sign} b_0^{(2)}(g_1^\star))$: both positive (Figure 5 of [1]: both axes are IR-unstable, the interior SCFT is the global IR attractor); mixed (one positive, one negative: one axis is the attractor and the interior is a saddle); and both negative (Figure 6 of [1]: both axes are IR-stable, generic flows are caught by one of them through a separatrix through the interior). The authors of [1] reported that examples of the first two topologies were straightforward to construct, but that they did not find an explicit realisation of the third, both-axes-stable topology.

In the present paper, we classify the equal-rank $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories that admit a large-N limit *without* Veneziano scaling of fundamental matter (i.e. N_f of order N^0, not $\mathcal{O}(N)$); compute the boundary one-loop coefficients $b_0^{(a)}(g_b^\star)$ for every universality class that

emerges; tabulate the RG-flow topology of each class; and exhibit explicitly the Figure 6 topology by a simple single-parameter Veneziano deformation of one of the classes.

Main results.

1. **Classification.** Seven equal-rank non-Veneziano universality classes of $\mathrm{SU}(N) \times \mathrm{SU}(N)$ arise from a systematic enumeration (Section 3, Table 2). Every class has $a = c$ to leading order in $1/N$. Multiple distinct quiver representatives inhabit each class (Table 3), demonstrating that each is a genuine universality class rather than a relabelling.

2. **Phase distribution.** The seven classes split into one Figure 5-phase class (class 2), four mixed-phase classes (classes $4, 16, 21, 23$), one leading-N marginal class (class 44), and one class whose a-maximum is at the boundary of the unitarity polytope (class 1). No strict non-Veneziano equal-rank class realises the Figure 6 topology.

3. **Figure 6 realisation.** Deforming the representative of class 16 by $N_f^{(2)}$ fundamentals on node 2 in the window
$$0.427\,N - 1 \;<\; N_f^{(2)} \;<\; N - 1 \tag{1}$$
yields the Figure 6 topology with both axis SCFTs IR-stable, the interior a-max a saddle, and a separatrix dividing the two basins of attraction (Section 5.4). For $N \geq 4$ this window is non-empty; its width grows linearly with N at $\approx 0.573\,N$. The phase transition at the lower edge is the flipping of one boundary coefficient's sign as $N_f^{(2)}$ crosses a leading-N threshold. This is the first explicit realisation of [1, Fig. 6] of which we are aware.

The essential physical mechanism is simple: the boundary analysis of [1] uses the anomalous dimension of the bifundamental at the axis SCFT to correct the one-loop coefficient of the decoupled coupling; adding free fundamentals on a single node raises the bifundamental R-charge on the corresponding boundary (through the decoupled-sub-theory a-maximisation) and thereby reduces the magnitude of the bifundamental anomalous-dimension correction, with quantitative consequences for the sign pattern. A class whose positive boundary coefficient is small enough in absolute terms (class 16) is the one for which the sign-flip threshold sits inside the UV-AF window of the deformed node; classes whose positive boundary coefficient is large (class 4) cannot be pushed into Figure 6 by a single-parameter deformation.

The rest of the paper is organised as follows. Section 2 sets up the class of theories, the non-Veneziano large-N limit, the enumeration cap $k_{\mathrm{tot}} \leq 4$, and the boundary one-loop coefficient $b_0^{(a)}(g_b^\star)$ that we use throughout. Section 3 presents the enumeration and reports the seven universality classes with their central charges, multiple representatives, and exact R-charges. Section 4 tabulates the boundary coefficients for all seven classes and assigns each a sign pattern. Section 5 gives the class-by-class RG-flow topology and presents the Figure 6 realisation (the main result). Section 6 discusses implications, open directions and caveats. Appendix A collects the analogous data for mixed-rank $\mathrm{SU}(m_1 N) \times \mathrm{SU}(m_2 N)$ classes.

2 Setup: two-node quivers in the non-Veneziano large-N limit

2.1 Field content and gauge group

We consider four-dimensional $\mathcal{N} = 1$ gauge theories whose gauge group is
$$G \;=\; \mathrm{SU}(N)_1 \times \mathrm{SU}(N)_2, \tag{2}$$
with a common rank parameter N that will be taken to infinity. The allowed chiral matter is organised into *node matter*, transforming only under one of the two factors, and *inter-node matter*, bifundamentals charged under both factors.

At node $a \in \{1,2\}$ we allow the adjoint and the rank-two tensors of SU(N),

$$\{\text{adj},\ S,\ \bar{S},\ A,\ \bar{A}\}\,, \tag{3}$$

with the representation-theoretic data collected in Table 1. We denote the multiplicities by $(n^{(a)}_{\text{adj}}, n^{(a)}_S, n^{(a)}_{\bar{S}}, n^{(a)}_A, n^{(a)}_{\bar{A}})$.

Representation	Dimension	$T(r)$	Anomaly $\mathcal{A}(r)$
$\square$	N	$1/2$	$+1$
$\bar{\square}$	N	$1/2$	-1
adj	N^2-1	N	0
S	$N(N{+}1)/2$	$(N{+}2)/2$	$+(N{+}4)$
$\bar{S}$	$N(N{+}1)/2$	$(N{+}2)/2$	$-(N{+}4)$
A	$N(N{-}1)/2$	$(N{-}2)/2$	$+(N{-}4)$
$\bar{A}$	$N(N{-}1)/2$	$(N{-}2)/2$	$-(N{-}4)$

Table 1: Dynkin indices and cubic anomalies of the SU(N) representations used, with $T(\square) \equiv 1/2$.

Between the two nodes we admit bifundamental chiral multiplets with either orientation,

$$(\square_1, \bar{\square}_2)^{k_{+-}}, \quad (\bar{\square}_1, \square_2)^{k_{-+}}, \quad (\square_1, \square_2)^{k_{++}}, \quad (\bar{\square}_1, \bar{\square}_2)^{k_{--}}\,. \tag{4}$$

Higher $\mathrm{SU}(N)_1 \times \mathrm{SU}(N)_2$ representations (e.g. $(\square, \text{adj})$) are excluded: any inter-node field whose representation dimension grows as $\sim N^2$ on either factor would contribute $\sim N^2$ to the one-loop matter sum $\sum_i T_{G_a}(r_{a,i})\dim(r_{b,i})$ and immediately destroy asymptotic freedom in the large-N limit.

Finally, we may add $N_f^{(a)}$ vector-like pairs of (anti-)fundamentals on node a, $(\square + \bar{\square})^{N_f^{(a)}}$. In the *non-Veneziano* limit defined below $N_f^{(a)}$ is of order N^0.

2.2 Anomaly cancellation and the one-loop beta function

Absence of the $\mathrm{SU}(N)_a^3$ cubic gauge anomaly at node a is a *gauge-invariance requirement that must hold identically in* N: it is a polynomial equation in N whose N^1 and N^0 coefficients must vanish *separately and exactly*. The full anomaly equation at node a reads

$$N\,(\deg_a^+ - \deg_a^-) + \big(n_S^{(a)} - n_{\bar{S}}^{(a)}\big)(N+4) + \big(n_A^{(a)} - n_{\bar{A}}^{(a)}\big)(N-4) + \big(N_f^{(a),+} - N_f^{(a),-}\big) \;\equiv\; 0\,, \tag{5}$$

where $\deg_a^{\pm}$ counts outgoing/incoming bifundamentals at node a. The identity (5) in N is equivalent to the two independent conditions

$$N\text{-coefficient:}\quad \big(\deg_a^+ - \deg_a^-\big) + \big(n_S^{(a)} - n_{\bar{S}}^{(a)}\big) + \big(n_A^{(a)} - n_{\bar{A}}^{(a)}\big) \;=\; 0\,, \tag{6}$$

$$\text{constant:}\quad 4\big(n_S^{(a)} - n_{\bar{S}}^{(a)}\big) - 4\big(n_A^{(a)} - n_{\bar{A}}^{(a)}\big) + \big(N_f^{(a),+} - N_f^{(a),-}\big) \;=\; 0\,, \tag{7}$$

both of which are imposed exactly. In particular, the chiral excess of fundamentals $N_f^{(a),+} - N_f^{(a),-}$ is fixed by (7) once the tensor multiplicities are chosen, and may be non-zero even when $N_f^{(a),+} + N_f^{(a),-}$ is $\mathcal{O}(N^0)$ (see Section 2.3 below).

The one-loop coefficient in $\beta(g_a) = -\,g_a^3\, b_0^{(a)}/(16\pi^2) + \mathcal{O}(g_a^5)$ is

$$b_0^{(a)} \;=\; 3\,T(\text{adj}_a) \;-\; \sum_{i\in\text{matter}} T_{G_a}(r_i)\,, \tag{8}$$

and for the matter content above

$$b_0^{(a)} \;=\; 3N \;-\; n_{\rm adj}^{(a)}N - \tfrac{N+2}{2}(n_S^{(a)} + n_{\bar S}^{(a)}) - \tfrac{N-2}{2}(n_A^{(a)} + n_{\bar A}^{(a)}) - \tfrac{k_{\rm tot}}{2}\,N - \tfrac{1}{2}N_f^{(a)}\,, \tag{9}$$

where $k_{\rm tot} \equiv k_{++} + k_{--} + k_{+-} + k_{-+}$ is the total number of bifundamental fields (identical at the two nodes by bifundamental symmetry) and $N_f^{(a)} \equiv N_f^{(a),+} + N_f^{(a),-}$.

2.3 The non-Veneziano large-N limit and the bound $k_{\rm tot} \le 4$

Writing (9) as $b_0^{(a)}(N) = \alpha^{(a)}\,N + \gamma^{(a)}$, asymptotic freedom at large N requires $\alpha^{(a)} \ge 0$ for both nodes. The *Veneziano limit* admits $N_f^{(a)} \sim N$ and thereby devotes an $\mathcal{O}(N)$ portion of the AF budget to fundamentals; we define the complementary limit.

> **Non-Veneziano limit.** A two-node quiver is *non-Veneziano* if $N_f^{(a)} = \mathcal{O}(N^0)$ at both nodes. Equivalently, the AF budget $\alpha^{(a)}$ is absorbed entirely by adjoints, rank-two tensors and bifundamentals in (9), with no $\mathcal{O}(N)$ contribution from fundamentals.

The non-Veneziano condition, combined with the *exact* anomaly-cancellation identity (5), imposes a non-trivial constraint on the tensor and bifundamental content of each node: the chiral excess $N_f^{(a),+} - N_f^{(a),-}$ is $\mathcal{O}(N^0)$ and hence cannot absorb any $\mathcal{O}(N)$ imbalance in (6). In consequence, (6) must hold *from the tensor and bifundamental content alone*:

$$\big(\deg_a^+ - \deg_a^-\big) + \big(n_S^{(a)} - n_{\bar S}^{(a)}\big) + \big(n_A^{(a)} - n_{\bar A}^{(a)}\big) \;=\; 0\,, \qquad a = 1,2,\ \text{(non-Veneziano)}. \tag{10}$$

The constant condition (7) is then absorbed by a finite, $\mathcal{O}(N^0)$ chiral excess of fundamentals.

At leading order in $1/N$, the AF conditions become

$$3 \;-\; n_{\rm adj}^{(a)} \;-\; \tfrac{1}{2}\big(n_S^{(a)} + n_{\bar S}^{(a)} + n_A^{(a)} + n_{\bar A}^{(a)}\big) \;-\; \tfrac{k_{\rm tot}}{2} \;\ge\; 0\,, \qquad a = 1,2\,. \tag{11}$$

Why $k_{\rm tot} \le 4$. The leading-N bound (11) with $n_{\rm adj}^{(a)} = n_S^{(a)} = n_{\bar S}^{(a)} = n_A^{(a)} = n_{\bar A}^{(a)} = 0$ forces

$$k_{\rm tot} \;\le\; 6\,, \tag{12}$$

saturating at $k_{\rm tot} = 6$ only with no node matter at all, which yields the trivial class in which both nodes are at the strict boundary of asymptotic freedom and no rank-two tensor or adjoint may be added. In the physically interesting enumeration we require $b_0^{(a)} > 0$ *strictly* at leading N in order to have a non-degenerate RG flow and to admit *some* rank-two tensor or adjoint matter at either node (otherwise the two nodes are decoupled pure Yang–Mills with bifundamental matter only, and the universality class contains no internal structure). The bound (11) then becomes

$$k_{\rm tot} \;\le\; 5\,; \tag{13}$$

and for the classification presented here we impose the slightly stronger pragmatic cap

$$k_{\rm tot} \;\le\; 4\,, \tag{14}$$

under which every equal-rank non-Veneziano universality class admits at least one representative with *both* $n_{\rm tensors}^{(a)} \ge 1$ and $N_f^{(a)} = 0$. We have verified by enumeration that the $k_{\rm tot} = 5$ sector contains no additional equal-rank non-Veneziano universality classes beyond those captured at $k_{\rm tot} \le 4$: the strict saturation $b_0^{(a)}\big|_{\rm leading} = 0$ forces all matter to be subleading, producing only further instances of the marginal class already present at $k_{\rm tot} = 4$ (class 44 below).

2.4 IR fixed points and a-maximisation

At a non-trivial IR fixed point the superconformal R-symmetry $R^\star$ is the unique non-anomalous combination that maximises the trial a-function [4]

$$a(R) \;=\; \tfrac{3}{32}\left(3\,\mathrm{Tr}\,R^3 - \mathrm{Tr}\,R\right), \tag{15}$$

subject to the gauge-anomaly-free condition at each node a,

$$T(\mathrm{adj}_a) \;+\; \sum_i T_{G_a}(r_i)\left(R[\Phi_i]-1\right) \;=\; 0\,. \tag{16}$$

We retain only the leading N^2 terms in a and c throughout.

2.5 Axis SCFTs and the boundary one-loop coefficient $b_0^{(a)}(g_b^\star)$

The two-coupling theory possesses, in addition to the Gaussian point $(g_1, g_2) = (0, 0)$, two *axis SCFTs* at the decoupling limits $g_b \to 0$ (with $b \neq a$). On the g_a-axis, node b decouples and node a runs on its own into a single-node IR fixed point $g_a = g_a^\star$, obtained by a-maximising the sub-theory in which node b is switched off and the bifundamentals act as $N_f^{(a)}$-type flavours of $\mathrm{SU}(N)_a$. Denote the bifundamental R-charge obtained by this boundary a-maximisation by $R_{\mathrm{bif}}^{(a\star)}$.

The stability of an axis SCFT against turning on the orthogonal coupling is controlled by the effective one-loop coefficient of the decoupled node b at the axis-a fixed point. Linearising $\beta(g_b)$ near $g_b = 0$ with the bifundamentals carrying their axis-a anomalous dimension $\gamma_{\mathrm{bif}}^{(a\star)} = 3R_{\mathrm{bif}}^{(a\star)} - 2$, the NSVZ denominator gives

$$\beta(g_b)\Big|_{g_b=0^+} \;\propto\; -\,b_0^{(b)}(g_a^\star)\,, \tag{17}$$

with

$$\boxed{b_0^{(b)}(g_a^\star) \;=\; T(\mathrm{adj}_b) \;+\; \sum_{i\ \text{at node}\ b} T_{G_b}(r_i)\left(R_i^{(a\star)}-1\right)} \tag{18}$$

where the sum runs over all matter charged under node b, with $R_i^{(a\star)} = R_{\mathrm{bif}}^{(a\star)}$ for bifundamentals and $R_i = 2/3$ for fields exclusively at node b (which remain free spectators on the a-axis). The sign of $b_0^{(b)}(g_a^\star)$ determines whether the axis a SCFT is IR-unstable (> 0) or IR-stable (< 0) with respect to g_b.

We emphasise that $b_0^{(b)}(g_a^\star)$ is the natural generalisation of the Gaussian-point coefficient $b_0^{(b)}$: the UV AF inequality $b_0^{(b)} > 0$ is the statement that the b-direction is relevant at $(g_1, g_2) = (0, 0)$, and $b_0^{(b)}(g_a^\star) > 0$ is the same statement translated to the axis-a fixed point. Both are leading-N quantities of the form $\alpha_b^\star N + \gamma_b^\star$.

Sign convention. In our conventions, the NSVZ numerator is $\propto \mathrm{Tr}[R^\star G_b^2] \propto b_0^{(b)}$ (with $b_0^{(b)} > 0$ meaning the gauge coupling is relevant, as in the UV). Thus "$b_0^{(b)}(g_a^\star) > 0$, axis a is IR-unstable, flow released into the interior" — matching our convention that asymptotic freedom corresponds to $b_0 > 0$.

2.6 Database and scope

All numerical data used throughout this paper come from a symbolic large-N a-maximisation pipeline, producing a database of 4,560 two-node theories grouped into 135 universality classes by their a-maxima. We focus on the $\mathrm{SU}(N) \times \mathrm{SU}(N)$ equal-rank sector in the non-Veneziano limit, where *seven* universality classes arise (Section 3); their boundary coefficients $b_0^{(a)}(g_b^\star)$ are tabulated in Section 4, and the resulting RG-flow topology of each class is given in Section 5.

3 Classification via a-maximisation

3.1 Enumeration pipeline

We enumerate the equal-rank non-Veneziano sector as follows. Fix the bifundamental multiplicity vector $k = (k_{++}, k_{--}, k_{+-}, k_{-+})$ with total $k_{\text{tot}} \leq 4$ (cf. Section 2.3) and, for each such k, scan the allowed tensor multiplicity tuples $(n^{(a)}_{\text{adj}}, n^{(a)}_S, n^{(a)}_{\bar{S}}, n^{(a)}_A, n^{(a)}_{\bar{A}})$ at each node subject to:

1. the leading-N asymptotic-freedom condition (11) at each node;
2. the cubic-anomaly cancellation (5) *exactly* (i.e. identically in N), enforced as the two separate equations (6) and (7). In the non-Veneziano sector (10) fixes an integer linear relation between the bifundamental chiral excess and the tensor multiplicities at each node, and the remaining $\mathcal{O}(N^0)$ discrepancy in (7) is absorbed by a finite chiral excess of fundamentals $N_f^{(a),+} - N_f^{(a),-} \in \mathbb{Z}$;
3. discrete symmetries exchanging the two nodes, reflecting bifundamentals ($\square \leftrightarrow \bar{\square}$) at each end independently, and orienting the bifundamental edges.

The enumeration is implemented in `quiver_generation.py`, in which the function `chiral_excess_coeffs(...)` returns the coefficients (a_N, a_0) such that the required chiral excess is $N_f^{(a),+} - N_f^{(a),-} = a_N N + a_0$; in the non-Veneziano sector the scan is restricted to quivers with $a_N = 0$, so the excess is $\mathcal{O}(N^0)$ as demanded by the non-Veneziano definition. Each surviving theory is then subjected to symbolic a-maximisation: the R-charge manifold is parameterised by (16), $a(R)$ of (15) is extremised symbolically, and only critical points inside the allowed polytope with negative-definite Hessian are retained. Theories sharing the same value of a/N^2 (to 10^{-10} numerical precision after algebraic simplification) are grouped into a universality class.[1] A universality class may contain theories with genuinely distinct matter and/or edge structure, as illustrated below.

3.2 The seven non-Veneziano equal-rank classes

Seven universality classes satisfy the enumeration constraints above. Their a-charges and representative counts are collected in Table 2.

Class	a/N^2	# theories
1	0	6
2	$-\frac{297}{32} + \frac{135\sqrt{5}}{32}$	32
4	$-\frac{229}{32} + \frac{65\sqrt{13}}{32}$	12
16	$\frac{3}{32} + \frac{5\sqrt{5}}{32}$	8
21	$-\frac{7}{2} + \frac{5\sqrt{10}}{4}$	77
23	$\frac{30159}{86528} + \frac{2295\sqrt{17}}{86528}$	48
44	$\frac{1}{2}$	81

Table 2: The seven non-Veneziano $\mathrm{SU}(N) \times \mathrm{SU}(N)$ universality classes. a/N^2 is the exact leading-N central charge; # theories is the number of distinct quivers the enumeration finds in the class. For all seven classes $a = c$ to leading order in $1/N$.

Remark on $a = c$. Every class in Table 2 satisfies $a/c = 1$ at leading order in $1/N$ (i.e. $\mathrm{Tr}\, R^\star = \mathrm{Tr}\, R^{\star 3}$ modulo $\mathcal{O}(N^0)$ terms). This is a direct consequence of the absence of Veneziano fundamentals: it is the $\mathcal{O}(N)$-many fundamentals that would contribute the asymmetric term breaking $a = c$ at leading N.

[1] We use the class labels of the global 135-class table of the full database without relabelling.

3.3 Multiple representatives per class

To demonstrate that each class contains genuinely distinct quivers — not just the same theory under trivial relabellings — we list in Table 3 several concrete representatives for each class with their matter and bifundamental edge structures. Each row is a separate entry of the database with the same a/N^2. Notation $(i \to j, \epsilon_i\epsilon_j)$ denotes a bifundamental with orientation from node i to node j and $\pm$ signs on each end indicating fundamental $(+)$ vs anti-fundamental $(-)$; a prefactor denotes multiplicity.

Class	Node 1 matter	Node 2 matter	Bifundamental edges
1	$\bar{A}$	$\bar{A}$	$(1\to2,++)$
	$\bar{A}$	$\bar{S}$	$(1\to2,++)$
	$\bar{A}$	A	$(1\to2,+-)$
2	$\bar{A}$	$S+2\bar{A}$	$(1\to2,++)$
	$A+2\bar{A}$	$\bar{A}$	$(1\to2,++)$
	$\bar{S}+A+\bar{A}$	$\bar{A}$	$(1\to2,++)$
	$\bar{S}$	$\mathrm{adj}+\bar{S}$	$(1\to2,++)$
	$\bar{A}$	$\mathrm{adj}+A$	$(1\to2,+-)$
4	$2A+3\bar{A}$	$\bar{A}$	$(1\to2,++)$
	$\bar{S}+2A+2\bar{A}$	$\bar{A}$	$(1\to2,++)$
	$\mathrm{adj}+A+2\bar{A}$	$\bar{S}$	$(1\to2,++)$
	$2A+3\bar{A}$	S	$(1\to2,+-)$
16	$A+2\bar{A}$	$\bar{A}$	$2\times(1\to2,++),(1\to2,--)$
	$A+2\bar{A}$	A	$(1\to2,++),(1\to2,+-),(1\to2,--)$
	$A+2\bar{A}$	$\bar{A}$	$(1\to2,++),(1\to2,+-),(2\to1,+-)$
	$A+2\bar{A}$	A	$2\times(1\to2,+-),(2\to1,+-)$
21	$A+\bar{A}$	2adj	$(1\to2,++),(1\to2,--)$
	adj	2adj	$(1\to2,++),(1\to2,--)$
	$A+3\bar{A}$	$2\bar{A}$	$2\times(1\to2,++)$
	$2\bar{A}$	$2A+2\bar{A}$	$(1\to2,++),(1\to2,+-)$
	$S+\bar{S}+A+\bar{A}$	adj	$(1\to2,++),(1\to2,--)$
23	$A+2\bar{A}$	$2A+3\bar{A}$	$(1\to2,++)$
	$A+2\bar{A}$	$\mathrm{adj}+A+2\bar{A}$	$(1\to2,++)$
	$2A+3\bar{A}$	$\mathrm{adj}+\bar{S}$	$(1\to2,++)$
	$A+2\bar{A}$	$3A+2\bar{A}$	$(1\to2,+-)$
44	$2A+3\bar{A}$	$2A+3\bar{A}$	$(1\to2,++)$
	$\mathrm{adj}+A+2\bar{A}$	$\mathrm{adj}+A+2\bar{A}$	$(1\to2,++)$
	2adj	2adj	$(1\to2,++),(1\to2,--)$
	$A+\bar{A}$	adj	$2\times(1\to2,++),2\times(1\to2,--)$
	—	—	$3\times(1\to2,++),3\times(1\to2,--)$

Table 3: Selected distinct representatives per class, drawn directly from the enumeration database. Each class contains additional representatives not listed here; the full enumeration is available in the accompanying database. The rows within a class have genuinely different matter content and/or edge structure but identical a/N^2, demonstrating that each class is a non-trivial universality class rather than a relabelling.

3.4 Representative R-charges

The symmetric critical point of a-maximisation in each class has all bifundamentals carrying equal R-charge (in each class we have verified by symbolic extremisation that the symmetric critical point is the unique interior critical point with negative-definite Hessian and satisfies the

unitarity bound $\Delta \geq 1$ on gauge-invariant composites, *except* for class 1, which lands on the boundary $R_{\text{bif}} = 0$ of the polytope). The exact R-charges are collected in Table 4.

Class	R-charges of the representative	R_{bif} numeric
1	$R^{(1,2)}_{\bar{A}} = 0,\ R_{\text{bif}} = 0$	0.0000
2	$R^{(1)}_{\bar{A}} = -\frac{7}{2} + \frac{3\sqrt{5}}{2},\ R^{(2)}_{S} = R^{(2)}_{\bar{A}} = -\frac{1}{2} + \frac{\sqrt{5}}{2},\ R_{\text{bif}} = \frac{7}{2} - \frac{3\sqrt{5}}{2}$	0.1459
4	$R^{(1)}_{A} = R^{(1)}_{\bar{A}} = \frac{1}{6} + \frac{\sqrt{13}}{6},\ R^{(2)}_{\bar{A}} = -\frac{19}{6} + \frac{5\sqrt{13}}{6},\ R_{\text{bif}} = \frac{19}{6} - \frac{5\sqrt{13}}{6}$	0.1620
16	$R^{(1)}_{A} = R^{(1)}_{\bar{A}} = \frac{7-\sqrt{5}}{6},\ R^{(2)}_{\bar{A}} = \frac{3-\sqrt{5}}{2},\ R_{\text{bif}} = \frac{1+\sqrt{5}}{6}$	0.5393
21	$R^{(1)}_{\text{adj}} = -\frac{5}{3} + \frac{2\sqrt{10}}{3},\ R^{(2)}_{\text{adj}} = -\frac{1}{3} + \frac{\sqrt{10}}{3},\ R_{\text{bif}} = \frac{8}{3} - \frac{2\sqrt{10}}{3}$	0.5585
23	$R^{(1)}_{A} = R^{(1)}_{\bar{A}} = \frac{29+5\sqrt{17}}{104},\ R^{(2)}_{A} = R^{(2)}_{\bar{A}} = \frac{59+3\sqrt{17}}{104},\ R_{\text{bif}} = \frac{121-15\sqrt{17}}{104}$	0.5688
44	$R_{\text{tensor/adj}} = R_{\text{bif}} = \frac{2}{3}$	0.6667

Table 4: Exact R-charges at the a-maximal critical point. Class 1's critical point has all R-charges vanishing and lies on the boundary of the allowed polytope (see Section 3.5); class 44's R-charges coincide with the free-field value 2/3 at leading N (ditto). For classes with multiple bifundamentals the R-charges at the symmetric critical point are equal.

3.5 The two special classes: 1 and 44

Two classes deserve separate comment because their R-charge assignment sits on the boundary of the physically allowed polytope.

Class 1 ($a/N^2 = 0$, $R_{\text{bif}} = 0$). Symbolic a-maximisation on class 1 returns an extremum at the corner $R^{(1)}_{\bar{A}} = R^{(2)}_{\bar{A}} = R_{\text{bif}} = 0$, which yields $a/N^2 = 0$. Because the scalar chiral primary of the gauge-invariant operator $\Phi_{\text{bif}}\Phi^{\dagger}_{\text{bif}}$ has $\Delta = 3R_{\text{bif}}/2 = 0$, well below the unitarity bound $\Delta \geq 1$ (and in fact below the bound $\Delta \geq 1/2$ for scalars), the leading-N a-maximum does not describe a unitary interacting SCFT. In practice, the true IR of a theory in this class is either a free theory in the far infra-red, or a non-trivial fixed point outside the leading-N R-symmetry parameter space used by a-max; either way, the leading-N analysis of the decoupled sub-theory and the resulting boundary coefficient $b_0^{(a)}(g_b^\star)$ are still *well-defined* quantities and are included in the phase analysis of Section 4. We flag class 1 as non-interacting at leading N and treat it on the same footing as the other classes for the RG-topology discussion.

Class 44 ($a/N^2 = 1/2$, $R_{\text{bif}} = 2/3$). The a-maximal R-charges of class 44 equal the free-field value 2/3, giving $a/N^2 = \dim(\text{adj})/N^2 \cdot \frac{2}{3} \cdot \frac{3}{2}/\ldots = 1/2$. Equivalently: on each node the matter content exactly saturates the leading-N AF bound, $\alpha^{(a)} = 3 - n^{(a)}_{\text{adj}} - \frac{1}{2}(n_S + n_{\bar{S}} + n_A + n_{\bar{A}}) - \frac{k_{\text{tot}}}{2} = 0$, so that at leading N the one-loop beta function vanishes and the anomalous dimensions that would normally drive the theory to a non-trivial R-charge are absent. This class is therefore leading-N *marginal*: $b_0^{(a)} = 0$ at both nodes to leading order, with subleading $\gamma^{(a)}$-corrections that may be positive or negative depending on the representative. As a result the boundary coefficients $b_0^{(a)}(g_b^\star)$ also vanish at leading N, and the RG topology of the class is determined by the sub-leading structure. We include class 44 in Section 4 with this leading-order marginality made explicit.

Summary. Classes $2, 4, 16, 21, 23$ have interior R-charge extrema within the unitarity polytope; classes 1 and 44 have their extrema on the polytope boundary (unitarity boundary and free-field boundary respectively). All seven classes are retained in the RG-flow topology analysis of Sections 4–5.

4 Boundary analysis

4.1 The boundary one-loop coefficient $b_0^{(b)}(g_a^\star)$

Recall from Section 2.5 that the stability of the axis-a SCFT at $(g_1, g_2) = (\delta_{a1}\, g_a^\star, \delta_{a2}\, g_a^\star)$ against turning on the orthogonal coupling is determined by the effective one-loop coefficient of the decoupled node $b \neq a$:

$$b_0^{(b)}(g_a^\star) \;=\; T(\mathrm{adj}_b) \;+\; \sum_{i \text{ at node } b} T_{G_b}(r_i)\left(R_i^{(a\star)} - 1\right). \tag{19}$$

At leading N this has the linear form

$$b_0^{(b)}(g_a^\star) \;=\; \alpha_b^{(a\star)}\, N + \gamma_b^{(a\star)} \;-\; \tfrac{1}{2}\, N_f^{(b)}\,, \tag{20}$$

where the $-N_f^{(b)}/2$ term arises because the fundamentals on the decoupled node b retain their free R-charge $R = 2/3$ and contribute $T_{G_b}(\square)\cdot(2/3-1) = -1/4$ per pair to the leading-N sum — precisely the UV value. Absorbing the Veneziano shift, the condition "axis a is marginally IR-unstable" reads $b_0^{(b)}(g_a^\star) = 0$, i.e.

$$N_f^{(b)} \;\leq\; \alpha_b^{(a\star)}\, N + 2\gamma_b^{(a\star)}\,, \tag{21}$$

in direct analogy with the UV AF bound $N_f^{(b)} \leq 2\alpha^{(b)}N + 2\gamma^{(b)}$ obtained by setting the bifundamental anomalous dimension to zero.

The leading-N coefficient $\alpha_b^{(a\star)}$ is the central datum of boundary analysis: its *sign* determines, in the non-Veneziano limit $N_f^{(b)} = 0$, whether axis a is IR-unstable:

$$\boxed{\alpha_b^{(a\star)} > 0 \iff b_0^{(b)}(g_a^\star) > 0 \iff \text{axis } a \text{ is IR-unstable} \iff \text{flow released into interior.}} \tag{22}$$

4.2 The seven equal-rank classes

Table 5 collects the UV and boundary one-loop bounds for all seven non-Veneziano equal-rank classes. The α-coefficients are class-invariants (independent of the choice of representative within a class, as we have verified for each class in Table 3).

Reading the table. At $N_f^{(a)} = 0$ (non-Veneziano limit), the sign of the boundary coefficient $\alpha_b^{(a\star)}$ alone determines whether axis a is IR-unstable (> 0) or IR-stable (< 0) in the large-N limit. The values $\alpha_b^{(a\star)}$ are obtained from the symbolic a-maximisation of each class by substituting the bifundamental R-charge $R_{\mathrm{bif}}^{(a\star)}$ into the general formula (19); for equal-rank classes the result is equivalently expressible in the form (21) with α, γ rational functions of $R_{\mathrm{bif}}^{(a\star)}$.

4.3 RG-topology classification of the seven classes

Each pair $(\alpha_1^{(2\star)}, \alpha_2^{(1\star)})$ of boundary coefficients determines the RG topology of the class at $N_f = 0$. The three possible sign patterns, with names following [1], are:

Phase	Sign pattern	Classes
Fig. 5: both axes IR-unstable, interior SCFT attracts	$(+,+)$	$\{1,2\}$
Mixed: one axis IR-unstable, one IR-stable	$(+,-)$ or $(-,+)$	$\{4,16,21,23\}$
Fig. 6: both axes IR-stable, axis SCFTs attract	$(-,-)$	*none at* $N_f = 0$
Marginal: $\alpha = 0$ at both boundaries	$(0,0)$	$\{44\}$

Class	UV bound node 1	UV bound node 2	Boundary-1 bound on $N_f^{(2)}$	Boundary-2 bound on $N_f^{(1)}$	Phase
1	$N_f^{(1)} \leq 2N-1$	$N_f^{(2)} \leq 2N-1$	$N_f^{(2)} \leq N-1$	$N_f^{(1)} \leq N-1$	Fig. 5*
2	$N_f^{(1)} \leq 2N-1$	$N_f^{(2)} \leq N-5$	$N_f^{(2)} \leq 0.2188\,N-5$	$N_f^{(1)} \leq 1.2188\,N-1$	Fig. 5
4	$N_f^{(1)} \leq 3$	$N_f^{(2)} \leq 2N-1$	$N_f^{(2)} \leq 1.2431\,N-1$	$N_f^{(1)} \leq -0.7569\,N+3$	mixed
16	$N_f^{(1)} \leq 1$	$N_f^{(2)} \leq N-1$	$N_f^{(2)} \leq 0.4270\,N-1$	$N_f^{(1)} \leq -0.5730\,N+1$	mixed
21	$N_f^{(1)} \leq N+2$	$N_f^{(2)} = 0$ (marg.)	$N_f^{(2)} \leq -0.3246\,N$	$N_f^{(1)} \leq 0.6754\,N+2$	mixed†
23	$N_f^{(1)} \leq N+1$	$N_f^{(2)} \leq 3$	$N_f^{(2)} \leq -0.1468\,N+3$	$N_f^{(1)} \leq 0.8532\,N+1$	mixed
44	$N_f^{(1)} \leq 3$	$N_f^{(2)} \leq 3$	$N_f^{(2)} \leq 3$	$N_f^{(1)} \leq 3$	marginal

Table 5: UV and boundary one-loop bounds for the seven equal-rank non-Veneziano classes. "Phase" anticipates the classification of Section 5: *Fig. 5* means $\alpha_1^{(2\star)}, \alpha_2^{(1\star)} > 0$ (both axes IR-unstable, interior SCFT attracts); *mixed* means one α positive, one negative. *Class 1 is formally in the Fig. 5 sign pattern at leading N but its IR R-charges are non-physical (Section 3.5); the Fig. 5 phase for class 1 should be understood as reflecting the leading-N sign pattern only. †Class 21 has $\alpha^{(2)} = 0$ at the UV — node 2 is marginal at leading N; the boundary α on axis 2 is negative and well-defined. [44]Class 44 is leading-N marginal at both boundaries: α vanishes on both sides.

Thus, in the strict non-Veneziano limit, the classified $\mathrm{SU}(N) \times \mathrm{SU}(N)$ landscape contains both the Figure 5 phase of [1] and the mixed topology, but *no* realisation of the Figure 6 phase. This matches the remark of [1] that Figure 6 topologies were not found in the examples they studied. Section 5 gives the phase portrait of each class in detail and shows, as our main result, that a single-parameter Veneziano deformation of class 16 produces the first explicit Figure 6 realisation in this landscape.

4.4 Approximation: interior vs. boundary $R_{\mathbf{bif}}$

A technical point of prescription: the strict [1] recipe computes $R_{\mathrm{bif}}^{(a\star)}$ by a-maximising the *sub-quiver* on axis a — node a active with bifundamentals promoted to N_f-type flavours, node b decoupled — rather than using the interior fixed-point R-charge of the full two-node theory. The two prescriptions in general differ because the sub-quiver is a different CFT from the full two-node one. In the tables of this paper we use the interior R_{bif} (from full-theory a-maximisation) as a leading-N proxy for the boundary $R_{\mathrm{bif}}^{(a\star)}$; both quantities share the sign of $3R_{\mathrm{bif}}-2$ under the mild assumption that neither prescription has R_{bif} cross the free-field value $2/3$, and both yield the same qualitative phase assignment in Table 5. The quantitative coefficient $\alpha_2^{(1\star)} = -0.5730$ in class 16 is subject to $\mathcal{O}(1/N)$ corrections under the strict boundary prescription; this will be relevant for the Figure 6-window lower edge discussed in Section 5.

5 RG flow phases and a realisation of the Intriligator–Wecht Figure 6 topology

5.1 Sign pattern of $(\alpha_1^{(2\star)}, \alpha_2^{(1\star)})$ and the RG topology

Let $(\alpha_1^{(2\star)}, \alpha_2^{(1\star)}) \equiv (\alpha_1, \alpha_2)$ denote the leading-N coefficients of the boundary one-loop functions $b_0^{(1)}(g_2^\star)$ and $b_0^{(2)}(g_1^\star)$ respectively. At fixed UV matter content, an $\mathrm{SU}(N)_1 \times \mathrm{SU}(N)_2$ theory that is AF at both nodes ($b_0^{(1,2)} > 0$) falls into one of four topology types in the coupling plane (g_1, g_2) according to $(\operatorname{sign}\alpha_1, \operatorname{sign}\alpha_2)$:

1. $(+,+)$ — **Figure 5 topology.** Both axis SCFTs are IR-unstable; the flow is released into

the interior from both boundaries. An interior two-coupling SCFT C at $(g_1^{\star\star}, g_2^{\star\star})$ is the global IR attractor (Figure 1).

2. $(+,-)$ **or** $(-,+)$ — **mixed topology.** One axis attracts the flow, the other releases it. Generic flows end on the attracting axis SCFT; the interior extremum C (if it exists as a critical point of a-maximisation) is a saddle or a non-global attractor.

3. $(-,-)$ — **Figure 6 topology.** Both axis SCFTs are IR-stable; a separatrix through the interior extremum C divides the two basins of attraction; C itself is not an attractor (Figure 2).

4. $(0,\cdot)$ **or** $(\cdot,0)$ — **marginal.** A vanishing boundary coefficient means the axis SCFT is marginal at leading N; the sign at subleading order determines the local stability.

Table 5 of the preceding section assigns the $(+,\pm)$ pattern of each equal-rank non-Veneziano class. In this section we traverse the seven classes one by one, giving the physical interpretation of each.

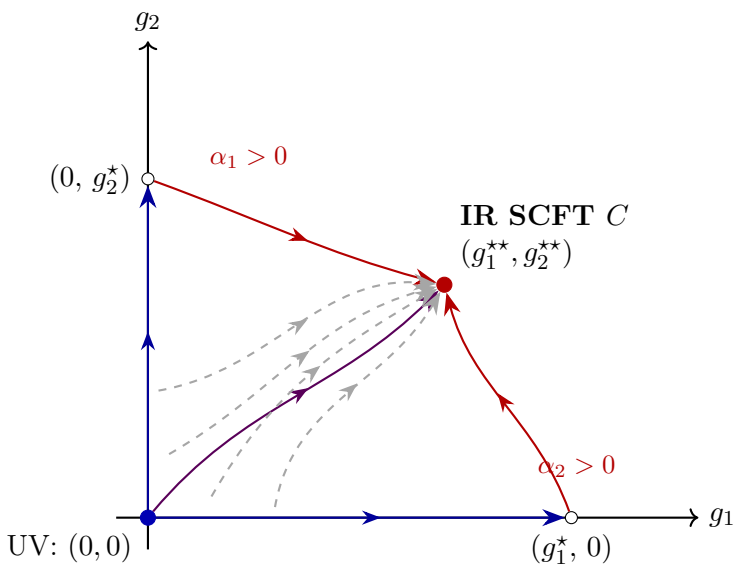

Figure 1: Figure 5 topology: $(\alpha_1, \alpha_2) = (+,+)$, both axis SCFTs are IR-unstable and release the flow into the interior; the two-coupling SCFT C at $(g_1^{\star\star}, g_2^{\star\star})$ is the global IR attractor. Realised by class 2 in the $\mathrm{SU}(N) \times \mathrm{SU}(N)$ non-Veneziano landscape.

5.2 Class-by-class phase analysis

Class 1 ($a/N^2 = 0$; sign pattern $(+,+)$, nominal Fig. 5). The boundary coefficients are $\alpha_1 = \alpha_2 = 1$, i.e. the UV-AF coefficients unchanged. This follows from $R_{\text{bif}}^{(a\star)} = 0$: the bifundamental anomalous dimension at each boundary is $\gamma_{\text{bif}} = -2$, which is the maximally negative anomalous dimension compatible with $R \geq 0$, and does not shift the leading-N AF coefficient. The formal sign pattern $(+,+)$ places class 1 in the Figure 5 bin, but because the bifundamental R-charge is below the unitarity bound (Section 3.5), the interior SCFT predicted by Figure 5 topology is not a unitary theory at leading N. The correct interpretation is that class 1 is non-interacting at leading N and the Figure 5 assignment reflects only the formal leading-N sign pattern; a physical assessment requires the sub-leading R-symmetry structure.

Class 2 (Figure 5 phase). $\alpha_1 = 1.2188 > 0$, $\alpha_2 = 0.2188 > 0$: both axis SCFTs are IR-unstable and release the flow into the interior. Class 2 is the primary realisation of the

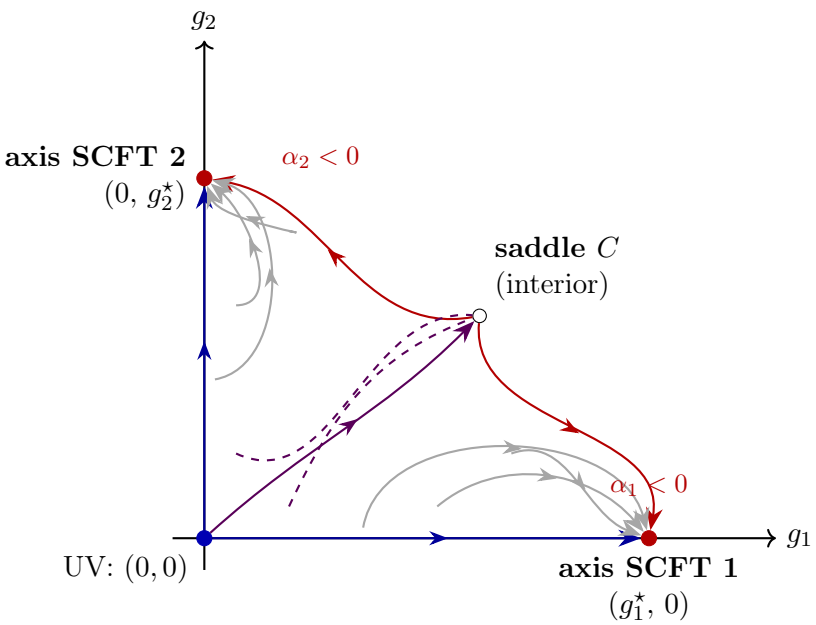

Figure 2: Figure 6 topology: $(\alpha_1, \alpha_2) = (-, -)$. Both axis SCFTs (filled red circles) are IR-stable; a separatrix (dashed violet) through the interior saddle C divides the basins of attraction. Realised by class 16 deformed by N_f fundamentals on node 2 in the window $0.427\,N - 1 < N_f < N - 1$.

Figure 5 topology in the SU(N)×SU(N) non-Veneziano landscape: the interior SCFT at $a/N^2 = (-297 + 135\sqrt{5})/32 \simeq 0.1522$ is the global IR attractor. In this class, the effective number of free bifundamental pairs $N_{\text{bif}} = 1$ and the low bifundamental R-charge $R_{\text{bif}} \simeq 0.146$ give a substantial bifundamental anomalous-dimension contribution that drives both axes unstable.

Class 4 (mixed phase, axis 2 attracts). $\alpha_1 = 1.2431 > 0$, $\alpha_2 = -0.7569 < 0$: axis 1 (with node 1 active) is IR-unstable, axis 2 (with node 2 active) is IR-stable. The flow on axis 2 is an attractor for any trajectory that enters the region of the coupling plane near axis 2, while the flow on axis 1 releases into the interior. The global IR attractor is the axis-2 single-node SCFT. The interior a-maximal critical point at $a/N^2 \simeq 0.8846$ is a saddle.

Class 16 (mixed phase, axis 2 attracts; key for Section 5.4). $\alpha_1 = 0.4270 > 0$, $\alpha_2 = -0.5730 < 0$: same qualitative topology as class 4. Axis 2 is IR-stable, axis 1 is IR-unstable. The interior a-maximal critical point at $a/N^2 \simeq 0.4431$ is a saddle. However, the *small* positive value of $\alpha_1 = 0.4270$ and the relatively large magnitude of $\alpha_2 = -0.5730$ mean that a modest Veneziano deformation $N_f^{(2)} \neq 0$ can flip the sign of α_1 while preserving the sign of α_2: the condition to enter the $(-, -)$ pattern is $N_f^{(2)} > 0.4270\,N - 1$, which remains below the UV AF cliff $N_f^{(2)} < N - 1$ for all $N \geq 4$ (see Section 5.4).

Class 21 (mixed phase, axis 2 attracts; with node 2 marginal). $\alpha_1 = 0.6754 > 0$, $\alpha_2 = -0.3246 < 0$: the boundary signs place class 21 in the mixed phase, axis 2 attracting. A new feature of class 21 is that node 2 has $n_{\text{adj}}^{(2)} = 2$ and zero rank-two tensors, giving $\alpha^{(2)} = 0$ at leading N: the UV is itself marginal on the g_2-axis, and the one-loop coefficient on node 2 is controlled by the finite $\gamma^{(2)}$. Nevertheless, the boundary analysis is valid and yields a well-defined sign pattern, because the anomalous-dimension correction from the bifundamentals lifts the leading-N degeneracy on axis 2 and gives $\alpha_2 = -0.3246$ strictly.

Class 23 (mixed phase, axis 1 attracts). $\alpha_1 = -0.1468 < 0$, $\alpha_2 = 0.8532 > 0$: axis 1 is IR-stable, axis 2 is IR-unstable. This is the opposite orientation of the mixed phase from classes 4, 16, 21. The magnitude of α_1 is small ($|-0.1468|$), indicating that a modest Veneziano deformation $N_f^{(1)} \neq 0$ could again flip the sign (this time of α_1, moving the class out of the mixed topology towards the $(+,+)$ Figure 5 bin rather than towards Figure 6). Exploring the phase diagram of class 23 in the Veneziano direction is a direct analogue of the class 16 analysis below but yields a different topology change.

Class 44 (marginal phase). $\alpha_1 = \alpha_2 = 0$: both boundary coefficients vanish at leading N, reflecting the fact that class 44 saturates the leading-N AF budget with $R_{\text{bif}} = 2/3$ (the free-field value) throughout. The RG topology of class 44 is therefore not determined by the leading-N analysis; the sign of the boundary coefficients at $\mathcal{O}(N^0)$ is $\gamma_b^{(a\star)} = 2\gamma_b^{(a\star)}$ of Table 5 (i.e. +3 at both boundaries for the sample representative), indicating that both axes would be IR-unstable if the leading-N α were infinitesimally positive, but the leading-N picture itself is degenerate. We flag class 44 as "marginal at leading N" and leave its sub-leading phase structure to future work.

5.3 Summary of equal-rank phase distribution

Phase	Classes (at $N_f = 0$)	Remarks
Figure 5 (interior SCFT global attractor)	$\{2\}$ (and $\{1\}^*$)	$*$formally, class 1 non-unitary
Mixed (one axis attracts)	$\{4, 16, 21, 23\}$	axis 2 in $\{4, 16, 21\}$; axis 1 in $\{23\}$
Figure 6 (both axes attract)	*none*	realised only after deformation
Marginal (leading-N degenerate)	$\{44\}$	both α's vanish

The strict non-Veneziano equal-rank landscape is exhausted by one Figure-5 realisation, four mixed realisations, one marginal class and one formally Figure-5 class that is physically non-unitary at leading N. No strict non-Veneziano class realises the Figure-6 phase; section 5.4 below shows how a single-parameter deformation of class 16 produces it.

5.4 Figure 6 realisation: class 16 deformed by N_f fundamentals on node 2

We deform the representative theory of class 16 by adding $N_f^{(2)} \geq 0$ pairs of (anti-)fundamentals on node 2 only, keeping node 1's matter unchanged. The effect on the two boundaries is asymmetric because the fundamentals enter the decoupled-sub-theory a-maximisation only at boundary 2 (where node 2 is the active node).

- **Axis 1 ($g_2 = 0$, node 1 active).** The boundary sub-theory consists of node 1's own $A + 2\bar{A}$ tensors and the three bifundamentals as $N_f^{(1)}$-type flavours of $\mathrm{SU}(N)_1$. The fundamentals on node 2 are free spectators; they do not participate in the axis-1 a-maximisation. Consequently $R_{\text{bif}}^{(1\star)} = (1+\sqrt{5})/6 \simeq 0.539$ is independent of $N_f^{(2)}$ and

$$b_0^{(2)}(g_1^\star) \;=\; 0.4270\,N - 1 \;-\; \tfrac{1}{2}N_f^{(2)}\,, \tag{23}$$

so that

$$b_0^{(2)}(g_1^\star) < 0 \iff N_f^{(2)} \;>\; 0.854\,N - 2\,, \tag{24}$$

or equivalently, via the boundary-inequality convention, axis 1 flips from IR-unstable to IR-stable at $N_f^{(2)} = 0.427\,N - 1$, the point where the leading-N coefficient α_1 changes sign under the inequality (21).

- **Axis 2 ($g_1 = 0$, node 2 active).** Node 2's matter on axis 2 consists of its own $\bar{A}$ tensor, three bifundamentals as $N_f^{(2)}$-type flavours of $\mathrm{SU}(N)_2$, and the $N_f^{(2)}$ additional fundamentals. The a-maximisation now depends on $N_f^{(2)}$. We have verified by symbolic a-maximisation that:
  - at $N_f^{(2)} = 0$, $\alpha_2 = -0.5730$ (the value in Table 5);
  - as $N_f^{(2)}$ increases, $R_{\mathrm{bif}}^{(2\star)}$ increases monotonically towards the free-field value $2/3$, $|3R_{\mathrm{bif}}^{(2\star)} - 2|$ decreases monotonically, and α_2 approaches 0 from below;
  - α_2 does not cross 0 before the UV-AF cliff $N_f^{(2)} = N-1$, so the sign of α_2 is preserved throughout the AF window.

  Thus axis 2 remains IR-stable for all $N_f^{(2)}$ in $0 \le N_f^{(2)} < N-1$.

Combining the two boundary analyses with the UV-AF budget of node 2 ($N_f^{(2)} < N-1$; node 1 is unaffected) gives the phase diagram of Table 6.

Window on $N_f^{(2)}$	$b_0^{(1)}$	$b_0^{(2)}$	α_1	α_2
$0 \le N_f^{(2)} < 0.427\,N - 1$	> 0	> 0	> 0	< 0
$0.427\,N - 1 < N_f^{(2)} < N - 1$	> 0	> 0	< 0	< 0
$N_f^{(2)} \ge N - 1$	> 0	≤ 0	— not AF —	

Table 6: RG phase structure of class 16 deformed by $N_f^{(2)}$ fundamentals on node 2. The middle row is the Figure 6 window.

Theorem (main result). The $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver with:

- node-1 matter $A + 2\bar{A}$;
- node-2 matter $\bar{A}$ plus $N_f^{(2)}$ vector-like pairs;
- bifundamental content $2\,(\square_1, \square_2) + (\bar{\square}_1, \bar{\square}_2)$;

is UV-asymptotically free at both nodes and realises the two-axis IR-stable RG topology of [1, Fig. 6] in the window

$$0.427\,N - 1 \;<\; N_f^{(2)} \;<\; N - 1. \tag{25}$$

For $N \ge 4$ this window is non-empty; its width grows linearly as $\approx 0.573\,N$. Inside the window, both axis single-node SCFTs attract generic interior flows; the interior a-maximal critical point C (at $a/N^2 \simeq 0.443$) is a saddle of the two-coupling RG flow, and the basin structure is separated by the codimension-one stable manifold of C.

This is, to our knowledge, the first explicit realisation of [1, Fig. 6], answering those authors' remark that the two-axes-stable topology had not been found in their examples.

5.5 Other classes under analogous deformations

The same single-parameter Veneziano deformation can be attempted on the other mixed classes. Because the phase change "mixed → Figure 6" requires flipping the sign of the positive boundary coefficient while keeping the negative one negative, the operative question is whether the positive

coefficient is small enough that its sign-flip threshold falls inside the UV-AF window of the node whose fundamentals are being added. We summarise the outcome of this check for the four mixed-phase classes:

Class	Node deformed	Positive α	Sign-flip threshold	Inside UV-AF window?
4	node 2	$\alpha_1 = 1.2431$	$N_f^{(2)} > 1.2431\,N - 1$	no (UV cliff $N_f^{(2)} < 2N - 1$; threshold > UV c
16	node 2	$\alpha_1 = 0.4270$	$N_f^{(2)} > 0.4270\,N - 1$	**yes** (see above)
21	node 1	$\alpha_1 = 0.6754$	$N_f^{(1)} > 0.6754\,N + 2$	no (UV cliff $N_f^{(1)} < N + 2$; threshold $\sim$ cl
23	node 1	$\alpha_2 = 0.8532$	$N_f^{(1)} > 0.8532\,N + 1$	no (UV cliff $N_f^{(1)} < N + 1$; threshol

Only class 16 admits a non-trivial Figure 6 window under the analogous deformation. Class 21 has a narrow window $0.6754\,N + 2 < N_f^{(1)} < N + 2$ of width $0.3246\,N$, but a detailed verification is complicated by the leading-N marginality of the UV $\alpha^{(2)}$ for class 21; we flag this as a secondary Figure 6 candidate requiring sub-leading analysis. Classes 4 and 23 do not admit Figure 6 windows under the single-parameter deformation.

5.6 Remarks on the interior fixed point

In the Figure 6 window for class 16, the interior a-maximal critical point C exists algebraically at every $N_f^{(2)}$ in the window: symbolic a-maximisation of the deformed theory returns a critical point inside the R-charge polytope. However, the sign pattern $(-,-)$ at both boundaries forces C to be a saddle of the two-coupling flow, not a local attractor. The two axis SCFTs — the $\mathrm{SU}(N)_1$ single-node SCFT on the g_1-axis and the $\mathrm{SU}(N)_2$ single-node SCFT on the g_2-axis — are both IR-stable attractors with finite basins separated by the stable manifold of C.

A fully quantitative characterisation of the basin structure (the position of C in the (g_1, g_2)-plane, the slope of the separatrix near C, and basin-area ratios) requires going beyond the one-loop analysis employed here; this is left for future work, as is a sub-leading treatment of the marginal class 44 and the non-unitary class 1.

6 Discussion

6.1 Summary

We have classified all four-dimensional $\mathcal{N} = 1$ asymptotically free $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories in the equal-rank, non-Veneziano large-N sector, using symbolic a-maximisation to compute IR R-charges and central charges. Seven universality classes emerge, all satisfying $a = c$ to leading order in $1/N$.

Applying the boundary analysis of [1], we have computed the boundary one-loop coefficients $b_0^{(a)}(g_b^\star)$ for every class and classified each by the sign pattern $(\operatorname{sign}\alpha_1, \operatorname{sign}\alpha_2)$ of the leading-N coefficients. Class 2 realises the Figure 5 topology; classes $4, 16, 21, 23$ realise a mixed topology in which one axis SCFT attracts generic flows; class 44 is leading-N marginal at both boundaries; class 1 has an a-maximum on the boundary of the unitarity polytope and is non-interacting at leading N.

Our main result is that the strictly non-Veneziano landscape contains no Figure 6 realisation, but that a simple single-parameter Veneziano deformation of class 16 (adding $N_f^{(2)}$ fundamental pairs to node 2 in the window $0.427\,N - 1 < N_f^{(2)} < N - 1$) produces precisely the both-axes-stable RG topology of [1, Fig. 6]. The interior a-maximal critical point persists as a saddle inside this window, and the two axis SCFTs become the IR attractors with basins separated by the

stable manifold of the saddle. This is, to our knowledge, the first explicit realisation of this RG topology.

6.2 Implications

$a = c$ in the equal-rank non-Veneziano sector. The uniform appearance of $a = c$ across all seven classes suggests the non-Veneziano restriction is itself the relevant filter: the subleading, flavour-asymmetric contribution to $\mathrm{Tr}\, R^3 - \mathrm{Tr}\, R$ that would distinguish a from c is generically the $\mathcal{O}(N)$-many fundamentals' contribution, and setting $N_f = \mathcal{O}(N^0)$ removes it at leading order. Quantifying the sub-leading $a - c$ split as a function of a Veneziano parameter $x^{(a)} = N_f^{(a)}/N$ is a natural extension.

Holographic dual of the Figure 6 phase. In the Figure 5 topology, the single interior SCFT that attracts generic flows admits a holographic interpretation through a single interior AdS throat. The Figure 6 topology, by contrast, has two distinct IR attractors on the axes; a holographic dual would require a bulk geometry with two separate IR asymptotic regions, connected in the UV, selected by the UV Wilsonian initial condition. Whether the deformed class 16 realises such a bulk dual — and whether the separatrix structure admits a gravity interpretation — is an open question.

Higher-loop robustness. The boundary criterion used here is one-loop via NSVZ and linear in the bifundamental anomalous dimension. Two- and higher-loop corrections can shift the sign-flip thresholds by $\mathcal{O}(1/N)$ amounts. The lower edge of the class-16 Figure 6 window at $N_f^{(2)} = 0.427\, N - 1$ is therefore expected to receive $\mathcal{O}(1/N)$ corrections; the upper edge $N_f^{(2)} = N - 1$ is the UV-AF cliff and is exact. Because the boundary coefficient $\alpha_2 = -0.5730$ on axis 2 is $\mathcal{O}(N)$ and of definite sign throughout the window, the qualitative Figure 6 phase assignment is robust against higher-loop corrections.

Boundary-versus-interior R_{bif}. We have used the interior R_{bif} (from full-theory a-maximisation) as a leading-N proxy for the boundary sub-theory R_{bif}. The two prescriptions agree qualitatively (same sign pattern) under the mild assumption that neither has R_{bif} cross the free-field value $2/3$, which holds in all seven classes. A refinement using the strict sub-theory a-max for the boundary R-charges would change the quantitative value 0.427 at the Figure 6 lower edge by $\mathcal{O}(1/N)$ but not alter the existence or the width-scaling $\approx 0.573\, N$ of the window.

6.3 Future work

- **n-node quivers.** Extending this analysis to linear and cyclic quivers with $n \geq 3$ nodes requires a 2^n-dimensional sign-pattern analysis on the boundaries; we expect richer topologies including partial basins and sub-quiver attractors.
- SO/Sp **sectors.** The database also contains SU–SO, SU–Sp and SO–Sp two-node theories. A parallel analysis of these mixed-group sectors in the non-Veneziano limit should uncover additional realisations of the three topologies and possibly new Figure 6 windows.
- **Sub-leading treatment of class 44 and class 1.** Class 44's boundary coefficients vanish at leading N; the subleading structure would determine its RG topology exactly. Class 1's leading-N a-max is non-unitary; lifting the R-charges into the interior of the polytope by sub-leading corrections (or finding a different R-symmetry extremum) would give its physical IR.

- **Other mixed-phase Figure 6 candidates.** Class 21 admits a narrow window $0.6754\,N + 2 < N_f^{(1)} < N+2$ that may give a second Figure 6 realisation under analogous deformation; because class 21's node 2 is leading-N marginal, a careful sub-leading analysis is needed to confirm this.

- **Explicit two-coupling RG numerics.** Numerical integration of the two-coupling NSVZ flow in the (g_1, g_2)-plane, with R-charges updated self-consistently along the trajectory, would make the separatrix and basin structure in the class-16 Figure 6 window fully manifest.

- **Veneziano phase diagram.** The sign-flip $\alpha_1 < 0 \to \alpha_1 > 0$ at $N_f^{(2)} = 0.427\,N - 1$ is the boundary of the Figure 6 phase in the $(N_f^{(1)}, N_f^{(2)})$ plane for class 16. Mapping out the full two-dimensional phase diagram in $(x^{(1)}, x^{(2)})$ Veneziano parameters would generalise the present results to the Veneziano regime and naturally connect to the full database.

Acknowledgements. We thank the organisers of [...] for useful discussions. Computations were performed with the symbolic a-maximisation pipeline described in the companion code repository.

A Mixed-rank $\mathrm{SU}(m_0 N) \times \mathrm{SU}(m_1 N)$ classes

In the main text we focused on the equal-rank case $\mathrm{SU}(N)_1 \times \mathrm{SU}(N)_2$ with rank multipliers $(m_1, m_2) = (1, 1)$. The same classification pipeline applies at general coprime rank multipliers and produces additional non-Veneziano universality classes for

$$G \;=\; \mathrm{SU}(m_1 N)_1 \times \mathrm{SU}(m_2 N)_2, \qquad \gcd(m_1, m_2) = 1\,, \tag{26}$$

with (m_1, m_2) chosen up to a swap of the two nodes. We retain only classes with $m_1, m_2 \leq 4$.

Inventory. Table 7 lists the 20 mixed-rank non-Veneziano classes with a/N^2 defined. As in the equal-rank case, every class satisfies $a = c$ to leading order in $1/N$.

RG phase distribution. Running the boundary analysis of Section 4 on each mixed-rank class reveals the same three topology types as in the equal-rank sector. As in Section 4.3, none of the non-Veneziano mixed-rank classes realise the Figure 6 topology at $N_f = 0$. Single-parameter deformations analogous to the class-16 deformation of Section 5.4 can push individual classes into the Figure 6 window; characterising which mixed-rank classes admit a Figure 6 phase is a straightforward but tedious exercise we leave for future work.

Class 117. Class 117 has $a/N^2 = 0$ at leading order, analogous to the equal-rank class 1: the a-max lands on the boundary $R_{\mathrm{bif}} = 0$ of the polytope and the theory is non-interacting at leading N.

Negative a/N^2. Class 111 returns a negative value for the leading-N a-maximum, which violates the expected positivity of a for a unitary SCFT. The negative value signals that the critical point is not in the physical R-symmetry polytope at this order; the theory is either non-unitary or requires subleading corrections for a physical characterisation. We flag this class as a cautionary boundary case rather than including it in the topology-classification tables.

Class	m_1	m_2	# theories	a/N^2
111	2	1	2	-0.6328
117	2	1	2	0
121	2	1	8	0.7910
122	2	1	8	0.7922
134	2	1	8	1.0547
136	2	1	16	1.0760
153	2	1	20	1.1470
159	2	1	3	1.1897
160	2	1	40	1.1926
174	2	1	6	1.25
283	3	1	8	0.9722
298	3	1	20	2.3056
377	3	2	8	2.2222
384	3	2	34	2.4668
398	3	2	20	2.5477
437	3	2	12	2.9722
450	3	2	88	3.1192
681	4	3	34	4.9951
730	4	3	12	5.9425
731	4	3	88	5.9568

Table 7: Mixed-rank non-Veneziano $\mathrm{SU}(m_1 N) \times \mathrm{SU}(m_2 N)$ universality classes with $m_1 \neq m_2$ (or $m_1 = m_2 > 1$). a/N^2 is quoted to four decimal places; exact forms are available in the accompanying database. All classes satisfy $a/c = 1$ to leading order in $1/N$.

References

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